

Solucions de la Fitxa 3: Radicals

Nota: en els apartats amb variables se suposa que aquestes són positives per poder simplificar sense valor absolut.

1. Calcular, sempre que es pugui, factoritzant prèviament el radicant i aplicant les propietats dels radicals

a) $\sqrt[3]{64} = 4$

b) $\sqrt{0,36} = 0,6 = \frac{3}{5}$

c) $\sqrt[3]{-8} = -2$

d) $\sqrt{0,01} = 0,1 = \frac{1}{10}$

e) $\sqrt[3]{-81} = -\sqrt[3]{81} = -\sqrt[3]{27 \cdot 3} = -3\sqrt[3]{3}$

f) $\sqrt[3]{1} = 1$

g) $\sqrt{-1}$ no té solució real.

h) $\sqrt[3]{-1} = -1$

i) $\sqrt[4]{(x^2)^3} = \sqrt[4]{x^6} = x^{6/4} = x^{3/2} = x\sqrt{x}$ (si l'enunciat real fos $((x^2)^3)^4$, aleshores el resultat seria x^{24}).

j) $\sqrt[3]{-32} = -\sqrt[3]{32} = -\sqrt[3]{8 \cdot 4} = -2\sqrt[3]{4}$

k) $\sqrt[3]{27} = 3$

l) $\sqrt{14400} = 120$

m) $\sqrt{225} = 15$

n) $\sqrt{1296} = 36$

o) $\sqrt{256} = 16$

p) $\sqrt{625} = 25$

q) $\sqrt{1024} = 32$

2. Expressar les potències en forma radical i viceversa

a) $5^{\frac{3}{4}} = \sqrt[4]{5^3} = \sqrt[4]{125}$

b) $3^{\frac{1}{2}} = \sqrt{3}$

c) $2^{\frac{5}{2}} = \sqrt{2^5} = \sqrt{32} = 4\sqrt{2}$

d) $11^{\frac{2}{5}} = \sqrt[5]{11^2} = \sqrt[5]{121}$

$$\text{e)} \quad \sqrt[3]{8^3} = 8^{\frac{3}{3}} = 8$$

$$\text{f)} \quad \sqrt[3]{9^2} = 9^{\frac{2}{3}} = (3^2)^{\frac{2}{3}} = 3^{\frac{4}{3}} = 3\sqrt[3]{3}$$

$$\text{g)} \quad \sqrt[3]{3} = 3^{\frac{1}{3}}$$

$$\text{h)} \quad \sqrt[3]{2} = 2^{\frac{1}{3}}$$

$$\text{i)} \quad \left(6^{\frac{3}{4}}\right)^{\frac{2}{5}} = 6^{\frac{3}{4} \cdot \frac{2}{5}} = 6^{\frac{3}{10}} = \sqrt[10]{6^3} = \sqrt[10]{216}$$

$$\text{j)} \quad \left(\left(\frac{2}{9}\right)^{\frac{1}{2}}\right)^{-1} = \left(\frac{2}{9}\right)^{-\frac{1}{2}} = \left(\frac{9}{2}\right)^{\frac{1}{2}} = \sqrt{\frac{9}{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$\text{k)} \quad \sqrt[3]{\sqrt[3]{2}} = \left(2^{\frac{1}{3}}\right)^{\frac{1}{3}} = 2^{\frac{1}{9}} = \sqrt[9]{2}$$

3. Extreure tots els factors que puguis del radical

$$\text{a)} \quad \sqrt{27} = \sqrt{9 \cdot 3} = 3\sqrt{3}$$

$$\text{b)} \quad \sqrt{60} = \sqrt{4 \cdot 15} = 2\sqrt{15}$$

$$\text{c)} \quad \sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$$

$$\text{d)} \quad \sqrt{180} = \sqrt{36 \cdot 5} = 6\sqrt{5}$$

$$\text{e)} \quad \sqrt{540} = \sqrt{36 \cdot 15} = 6\sqrt{15}$$

$$\text{f)} \quad \sqrt{243} = \sqrt{81 \cdot 3} = 9\sqrt{3}$$

$$\text{g)} \quad 3\sqrt{32} = 3\sqrt{16 \cdot 2} = 3 \cdot 4\sqrt{2} = 12\sqrt{2}$$

$$\text{h)} \quad 5\sqrt{27} = 5 \cdot 3\sqrt{3} = 15\sqrt{3}$$

$$\text{i)} \quad 4\sqrt{20} = 4 \cdot 2\sqrt{5} = 8\sqrt{5}$$

$$\text{j)} \quad \sqrt{189} = \sqrt{9 \cdot 21} = 3\sqrt{21}$$

$$\text{k)} \quad 2\sqrt{405} = 2\sqrt{81 \cdot 5} = 2 \cdot 9\sqrt{5} = 18\sqrt{5}$$

$$\text{l)} \quad \sqrt{2025} = 45$$

$$\text{m)} \quad \sqrt{\frac{2916}{64}} = \frac{\sqrt{2916}}{\sqrt{64}} = \frac{54}{8} = \frac{27}{4}$$

$$\text{n)} \quad \sqrt{\frac{135}{27}} = \sqrt{5} = \sqrt{5}$$

$$\text{o)} \quad \sqrt{6561} = 81$$

$$\text{p)} \quad \sqrt{2401} = 49$$

$$\text{q)} \quad \sqrt{256} = 16$$

- r) $\sqrt{625} = 25$
s) $\sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2}$
t) $\sqrt[3]{144} = \sqrt[3]{8 \cdot 18} = 2\sqrt[3]{18}$
u) $\sqrt[3]{28} = \sqrt[3]{28}$ (no té factors cúbics)
v) $\sqrt[3]{160} = \sqrt[3]{8 \cdot 20} = 2\sqrt[3]{20}$

4. Reduir a índex comú

- a) $\sqrt{5}$, $\sqrt[3]{4}$, $\sqrt[3]{6}$
m.c.m. dels índexs = 6:

$$\sqrt{5} = \sqrt[6]{5^3} = \sqrt[6]{125}, \quad \sqrt[3]{4} = \sqrt[6]{4^2} = \sqrt[6]{16}, \quad \sqrt[3]{6} = \sqrt[6]{6^2} = \sqrt[6]{36}.$$

- b) $\sqrt{32}$, $\sqrt[3]{10}$
m.c.m. dels índexs = 6:

$$\sqrt{32} = \sqrt[6]{32^3} = \sqrt[6]{32768}, \quad \sqrt[3]{10} = \sqrt[6]{10^2} = \sqrt[6]{100}.$$

- c) $\sqrt{3}$, $\sqrt[3]{4}$, $\sqrt[3]{8}$
m.c.m. dels índexs = 6:

$$\sqrt{3} = \sqrt[6]{3^3} = \sqrt[6]{27}, \quad \sqrt[3]{4} = \sqrt[6]{4^2} = \sqrt[6]{16}, \quad \sqrt[3]{8} = \sqrt[6]{8^2} = \sqrt[6]{64}.$$

- d) $\sqrt{xy}$, $\sqrt[3]{x^3}$, $\sqrt[3]{xy^6}$
m.c.m. dels índexs = 6:

$$\sqrt{xy} = \sqrt[6]{(xy)^3} = \sqrt[6]{x^3y^3}, \quad \sqrt[3]{x^3} = x = \sqrt[6]{x^6}, \quad \sqrt[3]{xy^6} = \sqrt[6]{x^2y^{12}}.$$

5. Fer les operacions, simplificant tant com es pugui

- a) $\sqrt{18} + \sqrt{50} - \sqrt{2} - \sqrt{8} = 3\sqrt{2} + 5\sqrt{2} - \sqrt{2} - 2\sqrt{2} = 5\sqrt{2}$
b) $\sqrt{243} + 3\sqrt{75} - 2\sqrt{147} = 9\sqrt{3} + 15\sqrt{3} - 14\sqrt{3} = 10\sqrt{3}$
c) $\sqrt{162x^3} - 4\sqrt{18x^5} - 5\sqrt{32x^7} = 9x\sqrt{2x} - 12x^2\sqrt{2x} - 20x^3\sqrt{2x} = (9x - 12x^2 - 20x^3)\sqrt{2x}$
d) $\sqrt{135} + 4\sqrt{40} - \sqrt{5^4} = 3\sqrt{15} + 8\sqrt{10} - 25$